

VEHICLE-CPS

Civil Protection Systems

A Projection-Governed Architecture for Environmental Safety, Perimeter Coherence and Controlled Human Response

Version 1.1 - Zenodo replacement draft

Environmental Safety through Relational Coherence

VEHICLE Systems Lab

Roberto Borda Milan

Santa Cruz, Bolivia

<https://vehiclesystemslab.com>

VEHICLE-CPS is not a surveillance system. It is a civil protection architecture for environmental safety, relational coherence and coordinated human response under mandatory human supervision.

Contents

1. Abstract
 2. Keywords
 3. Introduction
 4. Environmental Safety and Relational Coherence
 5. Borda Milan Pyramid Integration
 6. VEHICLE Mathematical Kernel
 7. Human Nodes and Operational Interfaces
 8. External Tension and Internal Incoherence
 9. Operational Regimes P0-P6
 10. Preventive Civil Protection Support
 11. Synthetic Simulation Phase: 1,000-10,000 Nodes
 12. KPIs and Evaluation Metrics
 13. Ethical Scope and Governance
 14. Partnership and Funding Pathway
 15. Limitations
 16. Conclusion
- Appendix A. Minimal Pseudocode

1. Abstract

VEHICLE-CPS - Civil Protection Systems is a civil protection research initiative by VEHICLE Systems Lab. It applies the Borda Milan Pyramid and the VEHICLE Formula-as-Architecture to high-density human environments such as public events, protected areas, stadiums, concerts, emergency drills, critical infrastructure environments and controlled response operations.

The central idea is that a mass human event is not only a crowd. It is a structured relational field under tension. People, authorized teams, smart bands, active credentials, access points, protected zones, evacuation routes, security rings and critical areas can be represented as nodes in a dynamic graph. These nodes carry structured E.I.A.R.(V) states and interact through relational tensions that may reveal loss of coherence before unsafe situations escalate.

VEHICLE-CPS is designed to strengthen environmental safety by helping authorized teams detect loss of relational coherence, rising pressure, route obstruction, perimeter deformation, disconnected operational nodes and controlled evacuation needs. It does not replace authorities, civil protection agencies, law enforcement, emergency services or private security companies. It provides an additional structural intelligence layer under mandatory human supervision.

The first development stage proposes synthetic simulations with 1,000 to 10,000 human nodes to validate computational feasibility, operational interpretation, ethical boundaries and institutional usefulness before any physical pilot. The project is positioned as preventive civil protection support, not as surveillance, profiling or autonomous enforcement.

2. Keywords

VEHICLE-CPS; Civil Protection Systems; Borda Milan Pyramid; VEHICLE Formula-as-Architecture; environmental safety; relational coherence; crowd safety; public events; perimeter coherence; controlled evacuation; structured nodes; E.I.A.R.(V); projection-governed correction; complex systems; human supervision.

3. Introduction

Modern public events are dynamic human systems. Thousands of people, access points, routes, operational teams, protected zones and emergency protocols interact under changing pressure. Risk does not always appear first as a visible incident. It may begin as a small loss of coherence: a blocked corridor, a broken security ring, an unexpected flow, a disconnected node, or a mismatch between where an operational credential should be and where presence is detected.

VEHICLE-CPS proposes a structural way to study this problem. Instead of focusing on isolated individuals, it reads the surrounding environment as a living relational field. The objective is to support authorized human teams in detecting early signs of environmental incoherence and responding faster, with clearer prioritization and better coordination.

The system is not designed to accuse, profile or identify attackers. It does not make coercive decisions automatically. It is an applied research architecture for civil protection, designed to help improve environmental safety through relational coherence and supervised decision support.

4. Environmental Safety and Relational Coherence

VEHICLE-CPS is designed to strengthen the safety of human environments by applying the Borda Milan Pyramid and the VEHICLE Formula-as-Architecture to civil protection scenarios. In this context, safety is understood as a property of the environment, not only as a property of individual persons.

A safe environment is one in which flows, access points, protected zones, routes, operational teams and verification mechanisms remain coherent under changing pressure. When coherence is lost, the system may move toward bottlenecks, perimeter pressure, evacuation stress, crowd surge or operational disconnection.

VEHICLE-CPS helps authorized teams read this loss of coherence before it becomes more difficult to control. It supports earlier verification, not automatic judgment. It improves the operational view of the environment, not the surveillance of identity.

5. Borda Milan Pyramid Integration

VEHICLE-CPS is an applied instance of the Borda Milan Pyramid. The pyramid provides an architectural pathway from an observed domain to relational modeling, tension measurement, projection-governed correction and decision support.

Pyramid layer	VEHICLE-CPS interpretation
Layer 1 - Observed domain	Public events, protected areas, stadiums, concerts, emergency drills and critical infrastructure environments.
Layer 2 - Relational conversion	The environment becomes a graph of people, teams, devices, access points, routes, protected zones and critical areas.
Layer 3 - Structured VEHICLE node	Each operational unit is represented through E.I.A.R.(V).
Layer 4 - Dual tension	External relational tension and internal incoherence are measured together.
Layer 5 - Projection-governed correction	The system supports admissible operational configurations.
Layer 6 - Operational regimes P0-P6	The environment is classified into civil protection states.
Layer 7 - Attractor discovery	Simulation reveals new patterns of crowd pressure, perimeter deformation and recovery.
Layer 8 - Decision support	Authorized teams receive structured information for faster verification and coordinated response.

6. VEHICLE Mathematical Kernel

The environment is represented as a graph:

$$G = (N, E)$$

Each node carries a structured state:

$$S_i = (E_i, I_i, A_i, R_i, V_i)$$

The total system tension is expressed as:

$$T(X) = T_{ext}(X) + T_{int}(X)$$

Projected correction follows the VEHICLE operator form:

$$V_op(S_i) = P_K[S_i - \gamma \text{grad}_{\{S_i\}} T(X)]$$

The relaxed update is:

$$S_i(t+1) = (1-\alpha)S_i(t) + \alpha V_op(S_i(t))$$

In VEHICLE-CPS, external tension may represent density, proximity, bottlenecks, perimeter breaks or route pressure. Internal incoherence may represent disconnected credentials, incompatible location-state relations, loss of signal or anomalous persistence. Projection-governed correction supports admissible operational response while leaving final interpretation and action under human supervision.

7. Human Nodes and Operational Interfaces

VEHICLE-CPS does not require that every person become an individually identified surveillance target. In its initial formulation, operational nodes may include authorized personnel, brigades, smart bands, active credentials, fixed tags, protected zones, access points, security rings, emergency routes and critical areas.

A smart band, active credential or equivalent device can act as the physical interface of a VEHICLE node. The system reads relational coherence between nodes, zones and routes rather than personal identity as such. In controlled simulations, synthetic nodes can represent general crowd behavior without collecting personal data.

8. External Tension and Internal Incoherence

External tension refers to relational pressure between nodes or zones. Examples include crowd density, unexpected flow, blocked routes, proximity stress, pressure on a security ring, or accumulation near critical access points.

Internal incoherence refers to a mismatch inside a node state. Examples include disconnected credentials, signal loss, incompatible location and authorization, abnormal persistence in a restricted area, or a critical operational team falling out of expected configuration.

VEHICLE-CPS combines these two dimensions to provide a more complete view of environmental safety. A crowd may appear calm but contain internal incoherence in critical nodes. Conversely, a visible concentration may be harmless if it remains structurally coherent and properly routed.

9. Operational Regimes P0-P6

VEHICLE-CPS introduces P0-P6 as an operational taxonomy for civil protection. These regimes translate relational tension into a language that can be understood by authorized teams, organizers, civil protection agencies and institutional partners.

Regime	Name	Meaning	Human action
P0	Structural Rest	Event not started or space essentially empty.	Normal readiness.
P1	Controlled Normal Flow	Occupancy and movement remain compatible.	Standard monitoring.
P2	Localized Review	Small anomalies	Nearby human check.

		requiring verification.	
P3	Anomalous Concentration	Bottlenecks, density pressure or unexpected crowd formation.	Redirect flow or reinforce sector.
P4	Perimeter Break / Critical Incoherence	Security ring breach, route conflict or critical node incoherence.	Immediate verification and containment.
P5	Reinforcement and Containment	Response is active and the field is being reconfigured.	Coordinated deployment.
P6	Controlled Evacuation	Partial or full evacuation under monitored tension.	Route protection and evacuation management.

10. Preventive Civil Protection Support

VEHICLE-CPS can support authorized human teams in situations involving perimeter pressure, anomalous concentration, route obstruction, crowd surge, unauthorized access patterns, disconnected operational nodes, evacuation stress, sudden changes in movement flow, high-emotion crowd behavior and coordination failures between security rings or response teams.

The system does not accuse, profile or identify attackers. It does not replace law enforcement, civil protection agencies or private security companies. It provides a structured decision-support layer that helps authorized human teams observe, verify and respond faster.

The core objective is prevention: reducing the time between the first signs of structural incoherence and coordinated human action.

11. Synthetic Simulation Phase: 1,000-10,000 Nodes

The first research phase of VEHICLE-CPS should be simulation-based. Before any physical pilot, the framework should be tested through synthetic human-node simulations at increasing scale: 1,000 nodes, 5,000 nodes and 10,000 nodes.

The simulation phase should test graph construction, tension computation, P0-P6 classification, local and global tension maps, detection timing, controlled response scenarios and recovery from high-tension states. This phase is designed to validate computational feasibility, operational interpretation, ethical boundaries and institutional usefulness.

12. KPIs and Evaluation Metrics

The following metrics are proposed for initial validation:

Metric	Meaning
Time-to-detection	Time between the beginning of a coherence loss and system classification.
Time-to-human-verification	Time between system alert and confirmation by authorized personnel.
Tension reduction	Change in T(X) after simulated response or route reconfiguration.
Return-to-coherence time	Time required to move from P4/P5 back toward

	P2/P1.
False escalation rate	Frequency with which local anomalies are incorrectly classified as systemic risk.
Route pressure index	Relative pressure on evacuation routes or access corridors.
Perimeter coherence score	Structural integrity of security rings or protected zones.

13. Ethical Scope and Governance

VEHICLE-CPS is designed around prevention, not persecution. Any physical pilot should require legal, ethical, operational and institutional review before deployment.

The ethical principles include mandatory human supervision, no automatic coercive action, data minimization, pseudonymized operational IDs, no facial recognition by design in the initial architecture, no political profiling, auditability of alerts, explicit limitation to civil protection and event safety use cases, and respect for human dignity.

14. Partnership and Funding Pathway

VEHICLE-CPS seeks strategic partners for simulation, validation and controlled civil protection research. The first development stage focuses on synthetic simulations of 1,000-10,000 human nodes to validate computational feasibility, P0-P6 classification, anomaly detection timing and controlled response scenarios before any physical pilot.

Potential partners include civil protection agencies, event security companies, municipalities, universities, emergency management institutions, wearable technology providers, venue operators and public safety researchers.

The initial partnership objective is not mass deployment. It is scientific, computational and institutional validation under clear ethical boundaries.

15. Limitations

VEHICLE-CPS is an applied research architecture, not a finished operational security product. It requires simulation, calibration, legal review, institutional validation and human-supervised pilot design before any real-world use.

The framework does not claim to predict individual intent. It does not replace human judgment. It does not eliminate risk. Its value lies in improving the structural reading of the environment and reducing the time between loss of coherence and coordinated human response.

16. Conclusion

VEHICLE-CPS extends the VEHICLE architecture into civil protection by treating high-density human environments as relational fields under tension. Through the Borda Milan Pyramid, it transforms observed public safety domains into structured graphs, measures external tension and internal incoherence, classifies operational regimes P0-P6 and supports admissible response under human supervision.

The project is best understood as environmental safety through relational coherence. It is not a surveillance system. It is a civil protection architecture for prevention, verification and coordinated response.

Appendix A. Minimal Pseudocode

Input: graph $G=(N,E)$, node states S_i , weights ω_{ij} , λ , step γ , relaxation α
for each timestep t :

```
    compute external tension  $T_{\text{ext}}$  over graph edges
    compute internal incoherence  $T_{\text{int}}$  for each structured node
     $T = T_{\text{ext}} + T_{\text{int}}$ 
    for each node  $i$ :
         $\text{grad}_i = \text{gradient of } T \text{ with respect to } S_i$ 
         $\text{candidate} = S_i - \gamma * \text{grad}_i$ 
         $\text{projected} = \text{project}_K(\text{candidate})$ 
         $S_{i_{\text{next}}} = (1-\alpha) * S_i + \alpha * \text{projected}$ 
    classify environment into P0-P6
    output tension map, regime state and human-verification recommendation
```

References and links

VEHICLE Systems Lab: <https://vehiclesystemslab.com>

GitHub repository: <https://github.com/vehiclesystemslab/VEHICLE-CPS>

Additional formal references and DOI metadata should be added after the new Zenodo version is created.